

Examining the Food Traceability trend through the use of Blockchain technology

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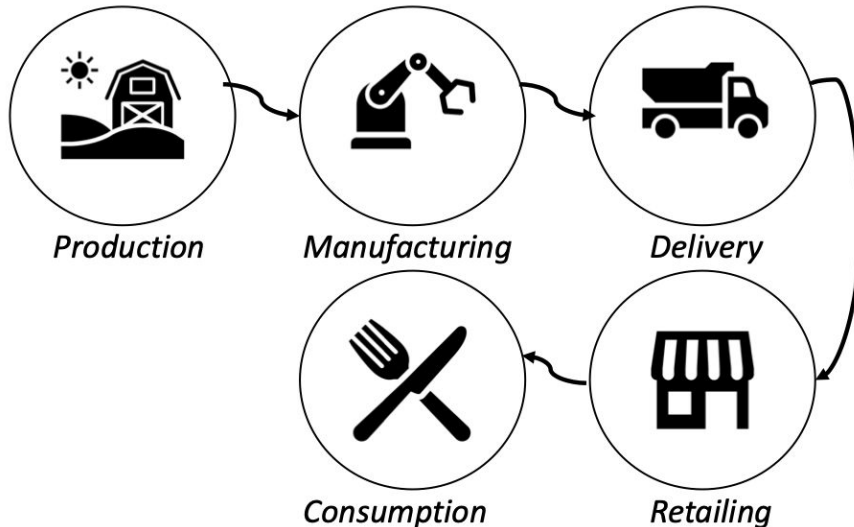
Content Page

- 1) What is Food Traceability?
- 2) What are common problems the industry faces and what are the corresponding gaps for innovation?
- 3) What is blockchain's role within the wider Food Traceability industry?
- 4) Case Study on the usage of blockchain in the context of Food Traceability
- 5) Long term implications & risks of blockchain with Food Traceability
- 6) Possible controversies

Food Traceability has been an emerging global trend underpinning food security

We first try to understand what Food Traceability is...

“ *The ability to track the origin and traces of a particular food product throughout its supply chain lifecycle.* ”



... and why it is important for global food security

1. Greater food safety for end-consumer

- ✓ *Faster identification of food contamination/spoilage*
- ✓ *More responsive and accurate product recalling*

2. Less food wastage throughout supply chain

- ✓ *With a faster call-to-action to stop production on defective products, lesser waste will be generated*
- ✓ *Promotes global food security through waste minimization*

Food Traceability as a trend has been plagued with problems of costs and data quality

Digesting the existing problems with food traceability...

1. Existing efforts are too expensive

- ✓ Clear lack of social/digital infrastructure in place
e.g., 44% of sub-Saharan Africa has a mobile device, and only 23% has access to mobile internet
- ✓ Existing legacy systems have no economies of scale for SMEs

2. Compromised data quality

- ✓ Current documentation efforts involve traditional avenues like paper documentation which jeopardizes data integrity
- ✓ Problem is compounded with growth of businesses' operations

3. Weak visibility across longer value chains

- ✓ With lack of data, difficult to see the big picture problem across entire food supply chain
- ✓ Less realistic for stakeholders to coordinate given siloed, and ineffective data

... allows us to better identify gaps for innovation

1. Cost efficiency over the long term



- ✓ Solution needs to be scalable, and cost-efficient over the long-term after infrastructure is in-place

2. Digitized documentation of data



- ✓ Data collected must be codified properly to ensure robust data analysis for food traceability actions

3. Robust participation from every stakeholder



- ✓ Food traceability must win through active participation of all users in the end-to-end supply chain process

How Blockchain is Utilized

- Blockchain technology can be utilized to trace a finished or marketable product back to its origin, farm to table.
 - The data is stored similarly to financial data on ledgers, the main use we have covered in class for this technology. on ledgers with specific QR codes relating to specific products and being tracked and uploaded as they make their way through the supply chain.
 - When BT is utilized with IoT, Internet of Things, the entire food supply chain can be tracked almost instantaneously.
- How it improves safety and efficiency.
 - Being able to pinpoint products exact location and path through the supply chain can help identify contaminated products in outbreaks and reduce waste in recalls when taking in to account affected products.
- Issues
 - Critics say blockchain can be a valuable piece of the food-safety puzzle but caution that it can easily be gamed. The online ledger requires manual entries, leaving it prone to human errors or intentional manipulation that could compromise the data chain.
 - The answer to this is putting more responsibility on the suppliers and by having them increase transparency through the use of BT can help achieve this increased responsibility and want to accurately monitor their food supply. meddling, health dangers aside, costs the food industry as much as \$49 billion a year.
 - Meddling and health dangers cost the food industry as much as 49 billion each year.

Who in the industry has begun using BT(blockchain technology)

Connecting food short video explanation for increasing customer transparency.

- <https://builtin.com/blockchain/food-safety-supply-chain>
- The use of this tracking system has already been demonstrated successful on chickens and the increased efficiency in the containment of salmonella outbreaks. Gogo chickens in CHina attaches QR lags to each chicken allowing for traceability from hatching to store.
- Bumblebee Foods, Tyson Foods, Kraft Heinz, Nestlé, and Walmart all are currently utilizing or testing out BT.

Will we see Governments adopting this system ?

- The USDA's Food Safety and Inspection Service has contracted with IBM to design a blockchain proof-of-concept project. This BT would allow better traceback and management of the FSIS food export certification project.

Wider adoption of food traceability on blockchain expected in the future, with certain risks

Less Developed Regions Left Behind



- Less developed regions **cannot afford such technology**
- **Widens disparity**

Mismatch between Technology & Needy Group



- **Children under 5 are at the highest risk** with 125,000 children dying every year from foodborne illness (*WHO*)
- **Children under 5 don't know blockchain**

Inherent Limitations Remain Unsolved



- **Participation from all parties involved and authentic reporting needed**
- Even right-minded parties **might report qualitative data differently** due to the lack of standards

Controversies over food traceability on blockchain should not be overlooked

Doesn't solve the root problem of food safety

- Blockchain enhances the ability to pinpoint potential sources of contamination to prevent, contain or rectify outbreaks
- **Fails to identify sources of food safety, e.g. cross-contamination**

Disadvantage small brands

- Transparency in terms of blockchain food traceability can validate and authenticate food origin and **improve brand credibility**
- **Small brands don't have the resources** to afford such technology are disadvantaged



Resources should be spent on food security

- 828M (**10%**) of the world's population are malnourished
- When so many people don't have enough food, it might be **better to direct resources on producing enough food** for them, then worrying about sub-standard foods



Room to cheat still exists

- Agents in the supply chain **enter information into a shared-based system**, which is unchangeable afterward
- **Room to cheat exists when information is inputted**

Sources

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